MATRICES

INVERSION

Inversion

A^{-1} – Inverse of matrix A

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad A^{-1} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}$$

determinant = ad – bc

if determinant = 0, then singular or noninvertible

Inversion

$$A = \begin{bmatrix} 10 & 2 \\ 4 & 1 \end{bmatrix}$$

determinant = 10*1 − 2*4 = 2

$$A^{-1} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -1 \\ -2 & 5 \end{bmatrix}$$